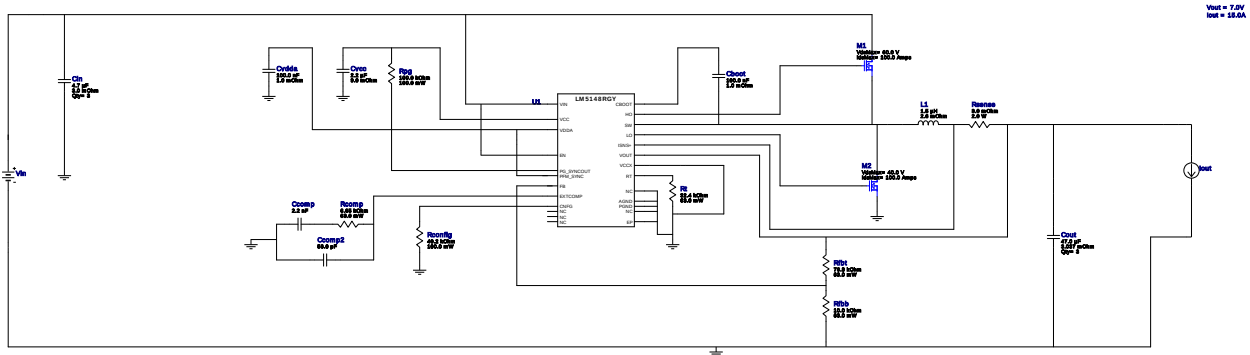


WEBENCH® Design Report

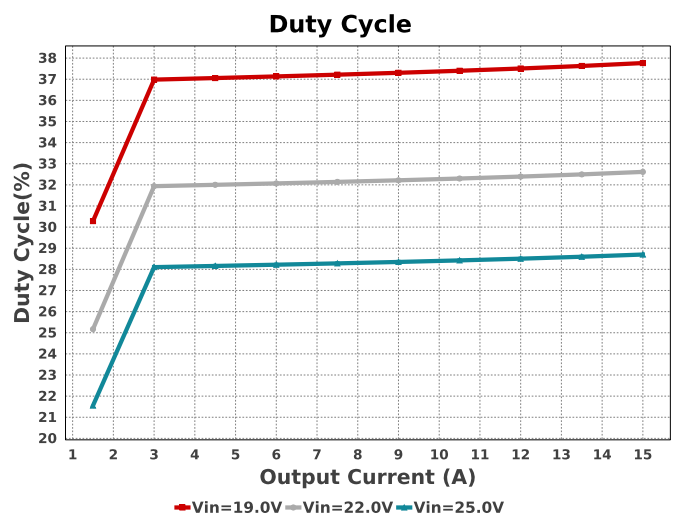
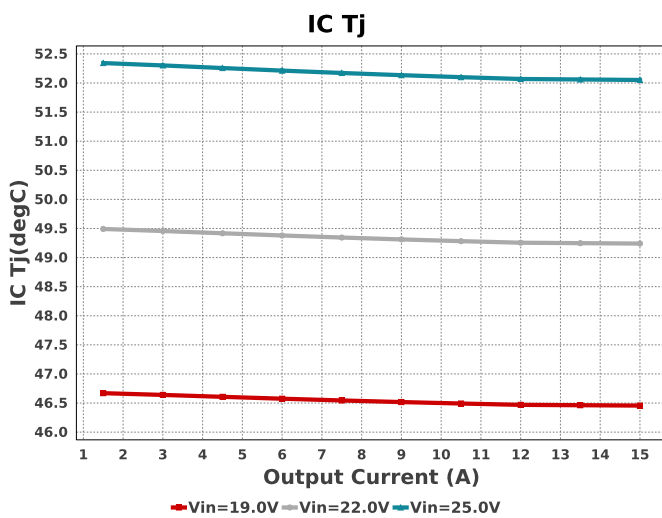
Design : 5 LM5148RGYR
LM5148RGYR 19V-25V to 10.00V @ 15A

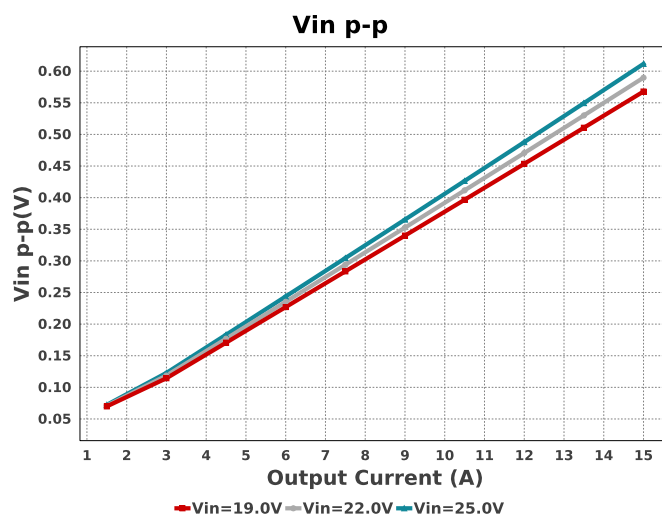
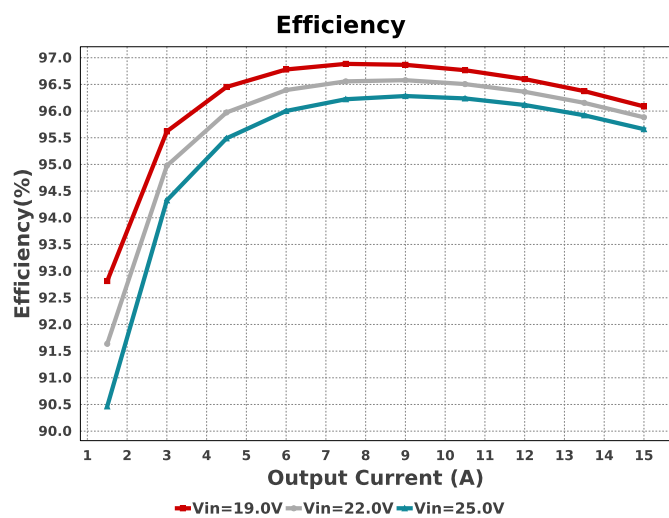
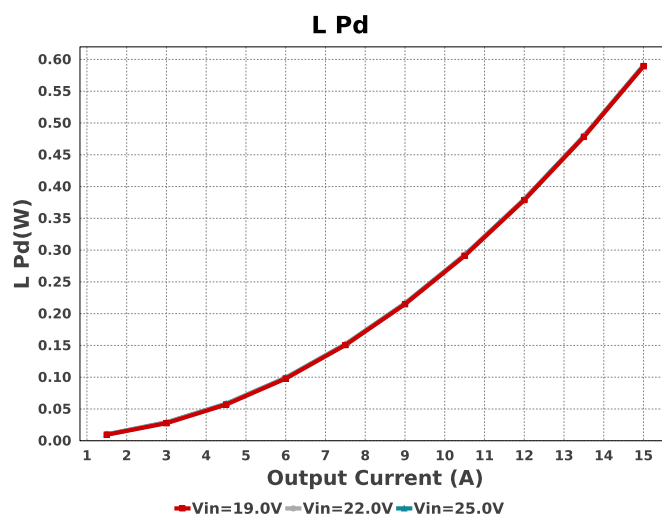
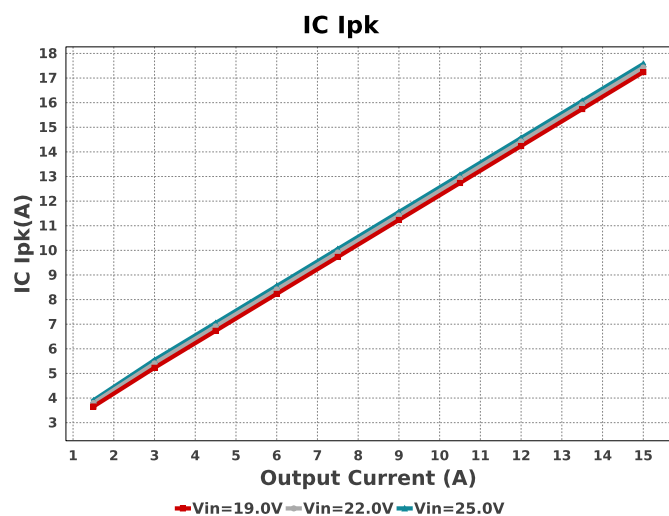
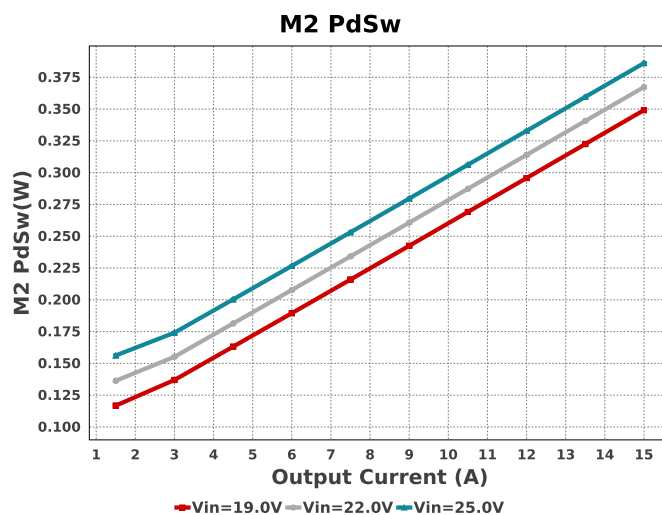
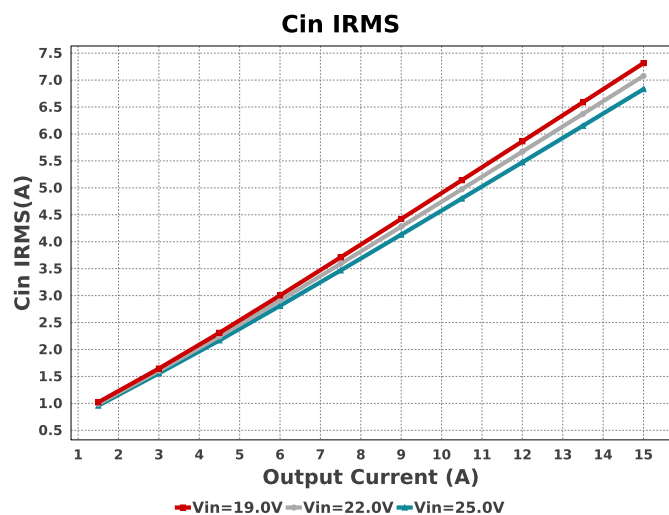


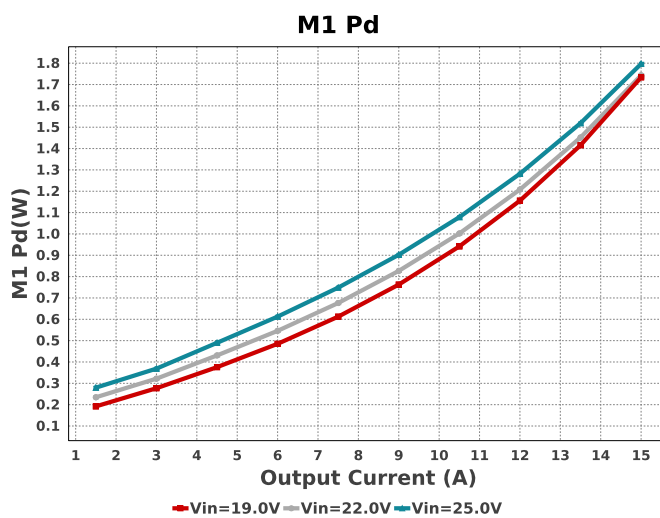
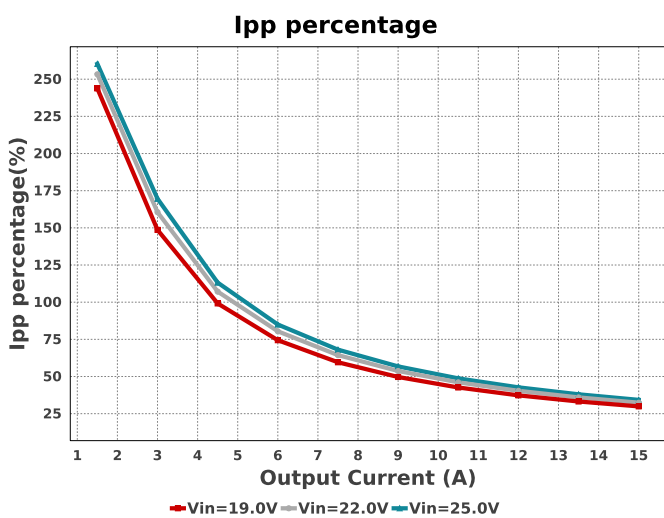
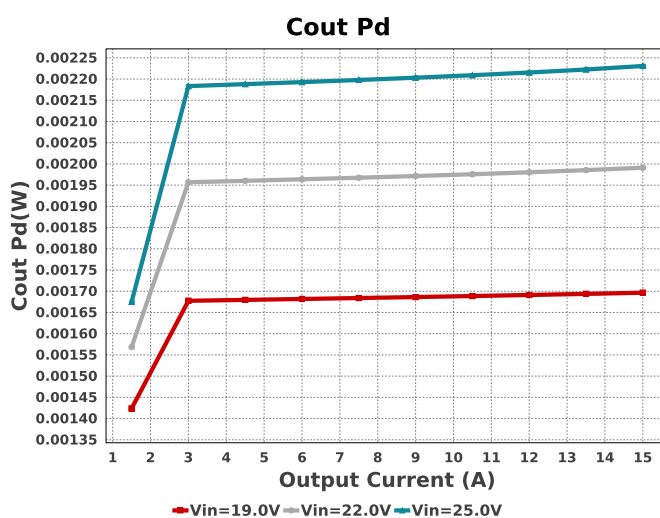
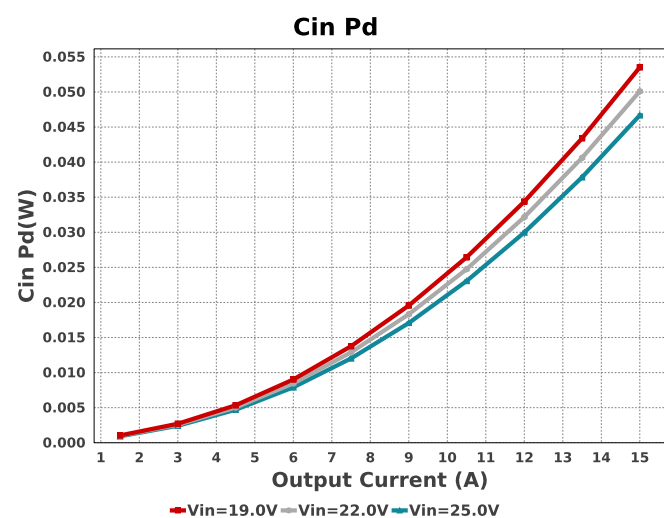
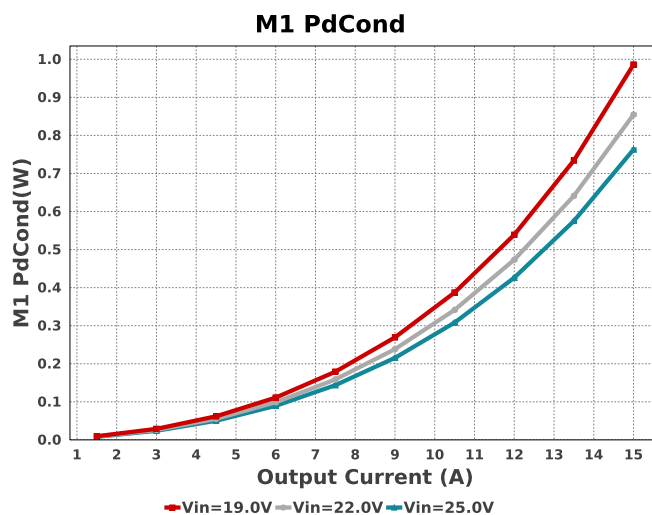
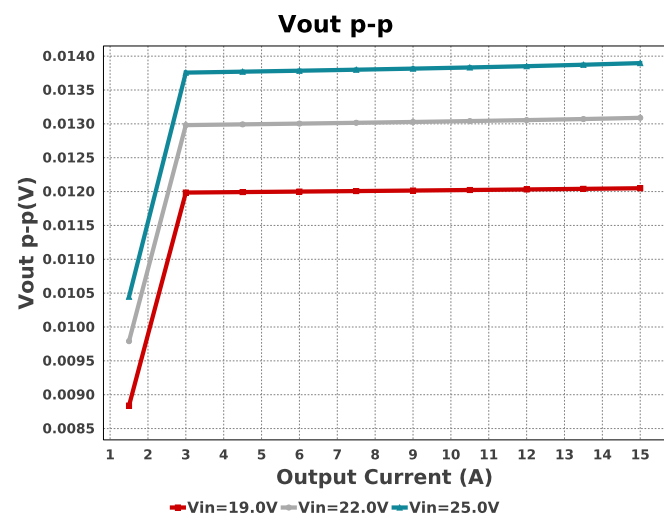
Electrical BOM

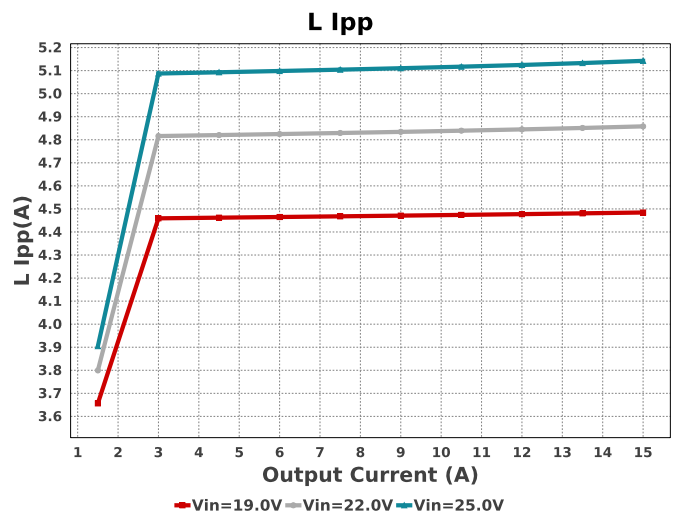
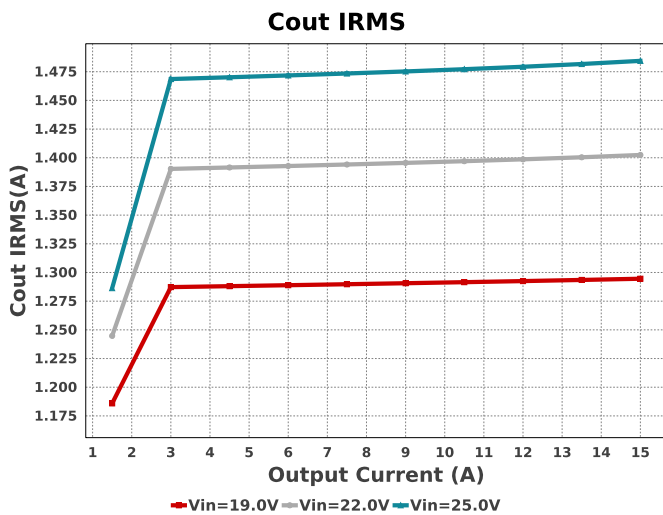
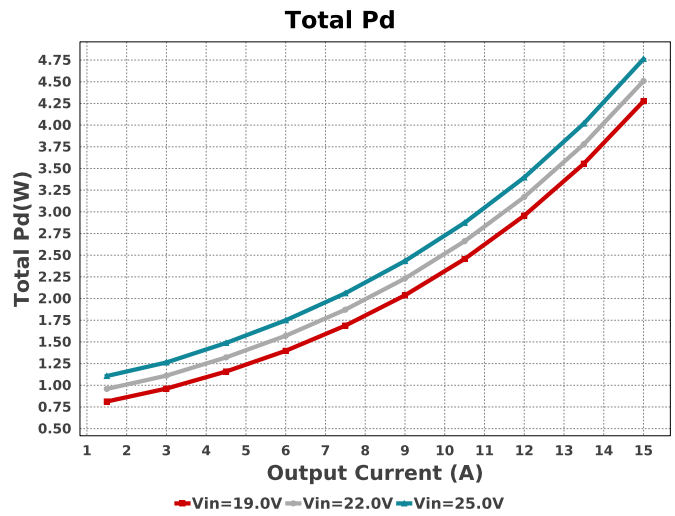
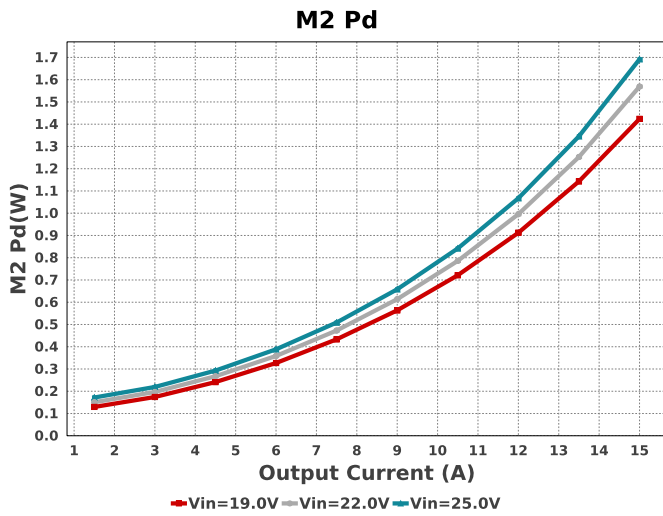
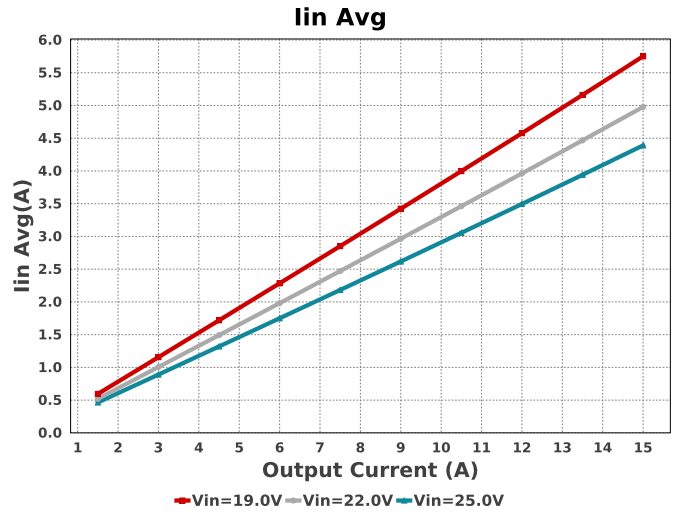
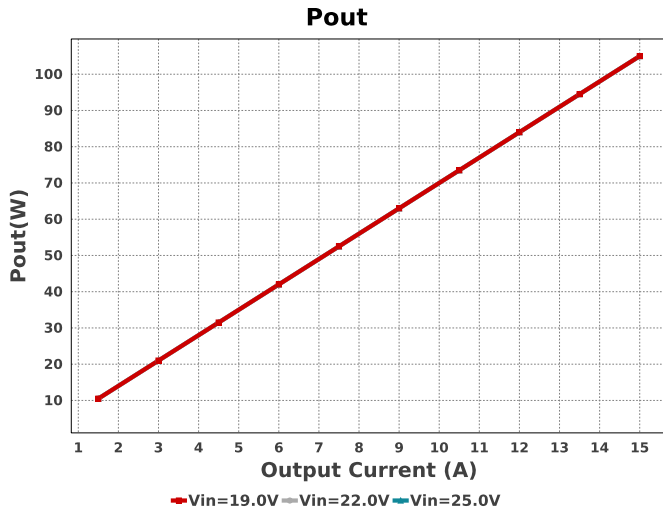
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cboot	MuRata	GRM155R71A104KA01D Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 10.0 V IRMS= 0.0 A	1	\$0.01	0402 3 mm ²
Ccomp	Samsung Electro-Mechanics	CL21C222JBFNNNE Series= C0G/NP0	Cap= 2.2 nF VDC= 50.0 V IRMS= 0.0 A	1	\$0.02	0805 7 mm ²
Ccomp2	Yageo	CC0805JRNPO9BN560 Series= C0G/NP0	Cap= 56.0 pF VDC= 50.0 V IRMS= 0.0 A	1	\$0.01	0805 7 mm ²
Cin	MuRata	GRM31CR71H475KA12L Series= X7R	Cap= 4.7 uF ESR= 3.0 mOhm VDC= 50.0 V IRMS= 4.98 A	3	\$0.10	1206 11 mm ²
Cout	MuRata	GRM32ER61C476KE15L Series= X5R	Cap= 47.0 uF ESR= 3.037 mOhm VDC= 16.0 V IRMS= 4.59346 A	3	\$0.17	1210_280 15 mm ²
Cvcc	MuRata	GRM188R71A225KE15D Series= X7R	Cap= 2.2 uF ESR= 9.0 mOhm VDC= 10.0 V IRMS= 3.3 A	1	\$0.02	0603 5 mm ²
Cvdda	MuRata	GRM155R71A104KA01D Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 10.0 V IRMS= 0.0 A	1	\$0.01	0402 3 mm ²
L1	Bourns	SRP1270-1R5M	L= 1.5 uH 2.6 mOhm	1	\$0.83	SRP1270 246 mm ²
M1	Texas Instruments	CSD18531Q5A	VdsMax= 60.0 V IdsMax= 100.0 Amps	1	\$0.45	TRANS_NexFET_Q5A 55 mm ²

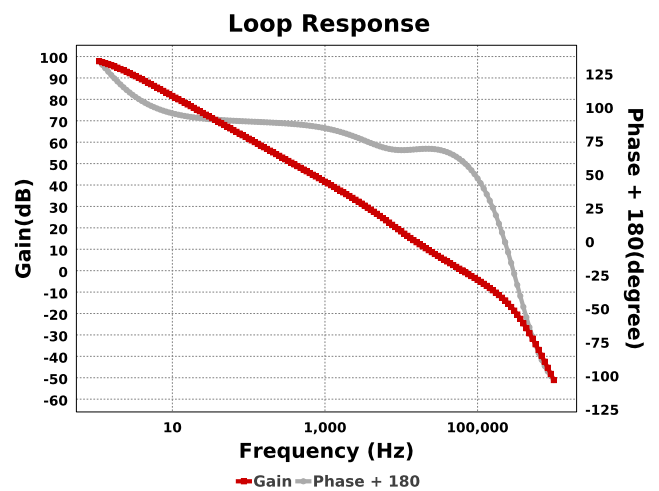
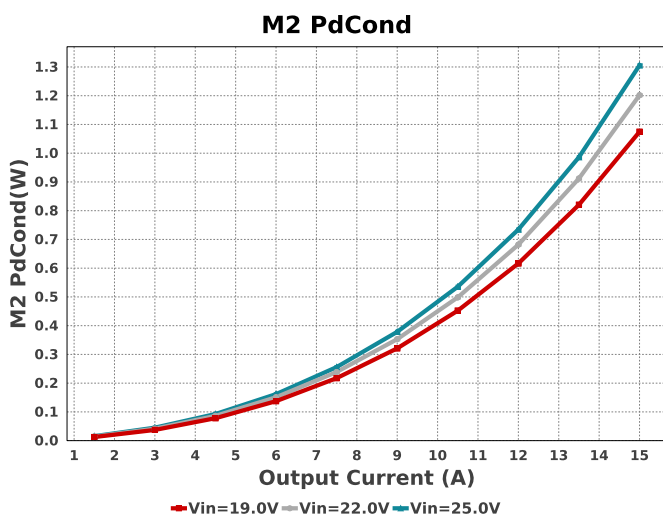
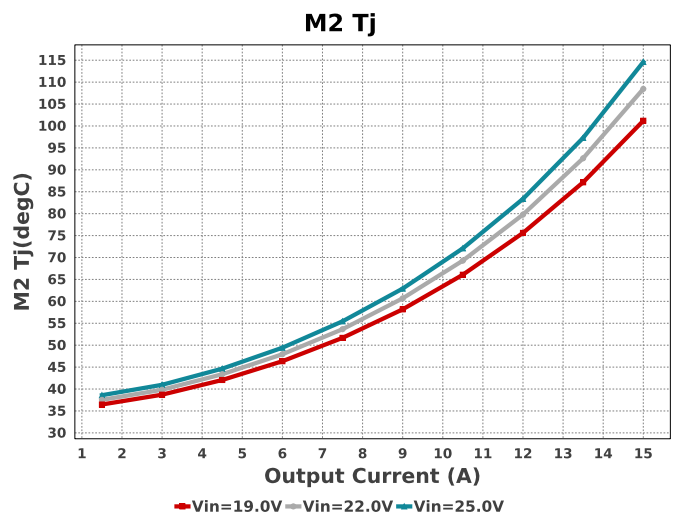
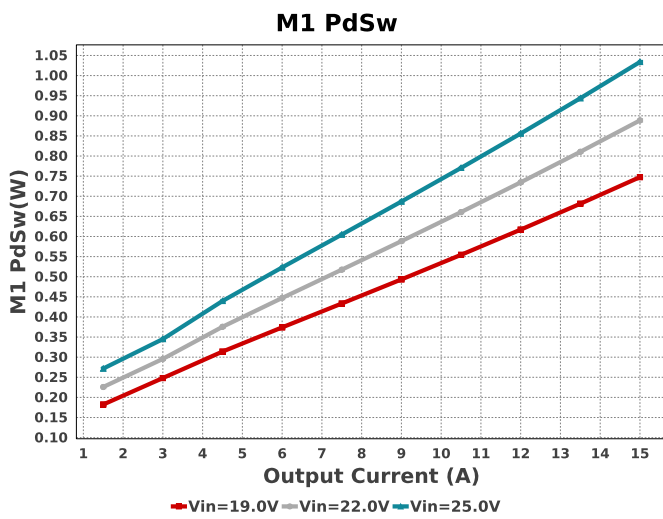
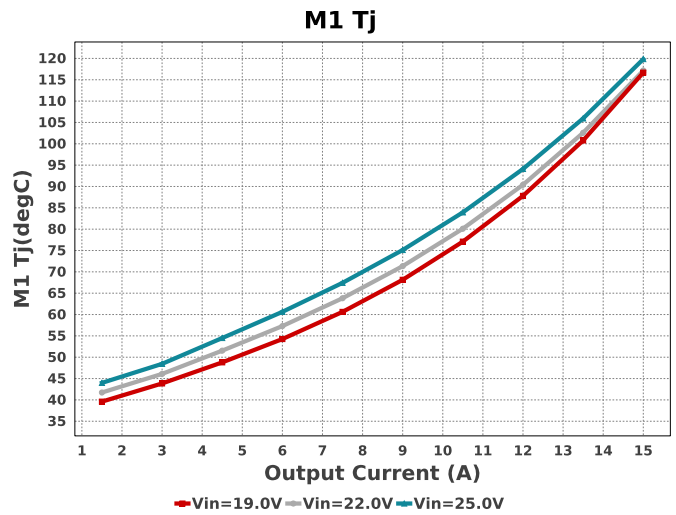
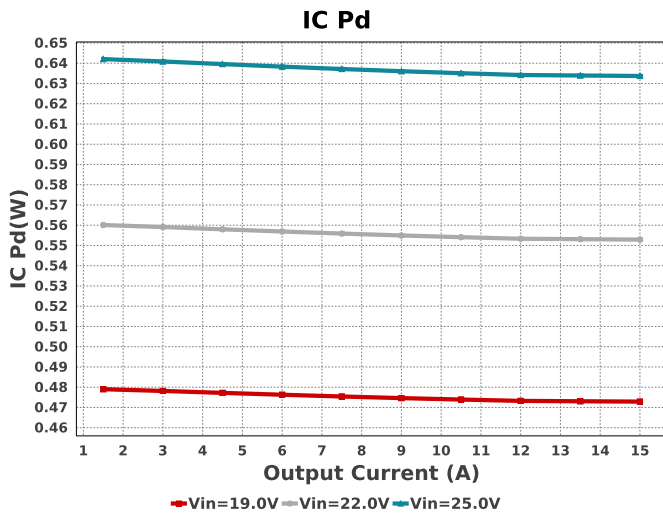
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
M2	Texas Instruments	CSD18513Q5A	VdsMax= 40.0 V IdsMax= 100.0 Amps	1	\$0.26	 TRANS_NexFET_Q5A 55 mm ²
Rcomp	Vishay-Dale	CRCW04026K65FKED Series= CRCW..e3	Res= 6.65 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
Rconfig	Vishay-Dale	CRCW060340K2FKEA Series= CRCW..e3	Res= 40.2 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	 0603 5 mm ²
Rfbb	Vishay-Dale	CRCW040210K0FKED Series= CRCW..e3	Res= 10.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
Rfbt	Vishay-Dale	CRCW040276K8FKED Series= CRCW..e3	Res= 76.8 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
Rpg	Vishay-Dale	CRCW0603100KFKEA Series= CRCW..e3	Res= 100.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	 0603 5 mm ²
Rsense	Vishay-Dale	WSR23L000FEA Series= WSR	Res= 3.0 mOhm Power= 2.0 W Tolerance= 1.0%	1	\$0.74	 4527 122 mm ²
Rt	Vishay-Dale	CRCW040232K4FKED Series= CRCW..e3	Res= 32.4 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
U1	Texas Instruments	LM5148RGYR	Switcher	1	\$0.60	RGY0024F 43 mm ²

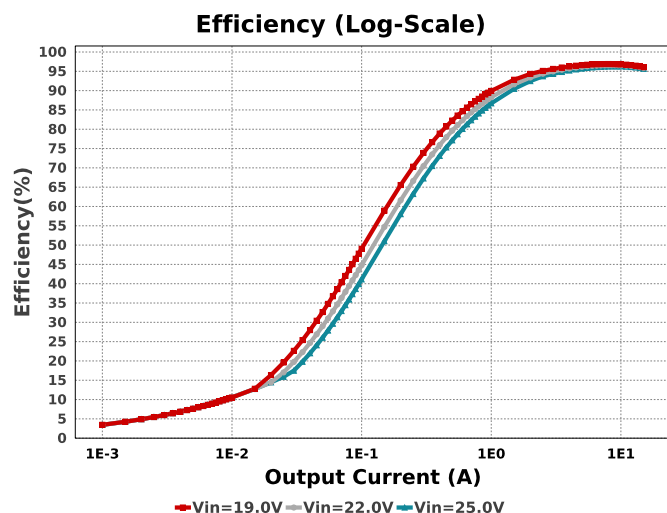












Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	6.832 A	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	46.677 mW	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	1.484 A	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	2.231 mW	Capacitor	Output capacitor power dissipation
5.	IC Ipk	17.571 A	IC	Peak switch current in IC
6.	IC Pd	633.69 mW	IC	IC power dissipation
7.	IC Tj	52.052 degC	IC	IC junction temperature
8.	IC Tolerance	10.0 mV	IC	IC Feedback Tolerance
9.	ICThetaJA Effective	34.8 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance
10.	Iin Avg	4.39 A	IC	Average input current
11.	Ipp percentage	34.282 %	Inductor	Inductor ripple current percentage (with respect to average inductor current)
12.	L Ipp	5.142 A	Inductor	Peak-to-peak inductor ripple current
13.	L Pd	590.73 mW	Inductor	Inductor power dissipation
14.	M1 Pd	1.797 W	Mosfet	M1 MOSFET total power dissipation
15.	M1 PdCond	763.0 mW	Mosfet	M1 MOSFET conduction losses
16.	M1 PdSw	1.034 W	Mosfet	M1 MOSFET switching losses
17.	M1 Tj	119.84 degC	Mosfet	M1 MOSFET junction temperature
18.	M2 Pd	1.692 W	Mosfet	M2 MOSFET total power dissipation
19.	M2 PdCond	1.306 W	Mosfet	M2 MOSFET conduction losses
20.	M2 PdSw	386.17 mW	Mosfet	M2 MOSFET switching losses
21.	M2 Tj	114.6 degC	Mosfet	M2 MOSFET junction temperature
22.	Cin Pd	46.677 mW	Power	Input capacitor power dissipation
23.	Cout Pd	2.231 mW	Power	Output capacitor power dissipation
24.	IC Pd	633.69 mW	Power	IC power dissipation
25.	L Pd	590.73 mW	Power	Inductor power dissipation
26.	M1 Pd	1.797 W	Power	M1 MOSFET total power dissipation
27.	M1 PdCond	763.0 mW	Power	M1 MOSFET conduction losses
28.	M1 PdSw	1.034 W	Power	M1 MOSFET switching losses
29.	M2 Pd	1.692 W	Power	M2 MOSFET total power dissipation
30.	M2 PdCond	1.306 W	Power	M2 MOSFET conduction losses
31.	M2 PdSw	386.17 mW	Power	M2 MOSFET switching losses
32.	Total Pd	4.763 W	Power	Total Power Dissipation
33.	BOM Count	22	System	Total Design BOM count
34.	Cross Freq	63.615 kHz	Information	Bode plot crossover frequency
35.	Duty Cycle	28.702 %	Information	Duty cycle
36.	Efficiency	95.661 %	Information	Steady state efficiency
37.	FootPrint	644.0 mm ²	Information	Total Foot Print Area of BOM components
38.	Frequency	661.813 kHz	Information	Switching frequency
39.	Gain Marg	-14.002 dB	Information	Bode Plot Gain Margin
40.	Iout	15.0 A	Information	Iout operating point
41.	Iout transient step used 7.5 A for Cout calculations		Information	Custom Transient current step requirement that was used for Cout selection (A).

#	Name	Value	Category	Description
42.	Low Freq Gain	97.817 dB	System Information	Gain at 1Hz
43.	Mode	CCM	System Information	Conduction Mode
44.	Overshoot Value	82.366 mV	System Information	Theoretical Vout Overshoot Value
45.	Phase Marg	60.427 deg	System Information	Bode Plot Phase Margin
46.	Pout	105.0 W	System Information	Total output power
47.	Total BOM	\$3.82	System Information	Total BOM Cost
48.	Undershoot Value	143.543 mV	System Information	Theoretical Vout Undershoot Value
49.	Vin	25.0 V	System Information	Vin operating point
50.	Vin p-p	611.554 mV	System Information	Peak-to-peak input voltage
51.	Vout	7.0 V	System Information	Operational Output Voltage
52.	Vout Actual	6.944 V	System Information	Vout Actual calculated based on selected voltage divider resistors
53.	Vout Ripple requirement used for Cout calculations	1.0 %	System Information	Custom maximum output ripple requirement that was used for Cout selection(% of Vout).
54.	Vout Tolerance	3.06 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
55.	Vout p-p	13.897 mV	System Information	Peak-to-peak output ripple voltage
56.	Vout transient requirement used for Cout calculations	3.0 %	System Information	Custom Transient voltage change requirement that was used for Cout selection (% of Vout).

Design Inputs

Name	Value	Description
Iout	15.0	Maximum Output Current
VinMax	25.0	Maximum input voltage
VinMin	19.0	Minimum input voltage
Vout	7.0	Output Voltage
base_pn	LM5148	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of $L1$ before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

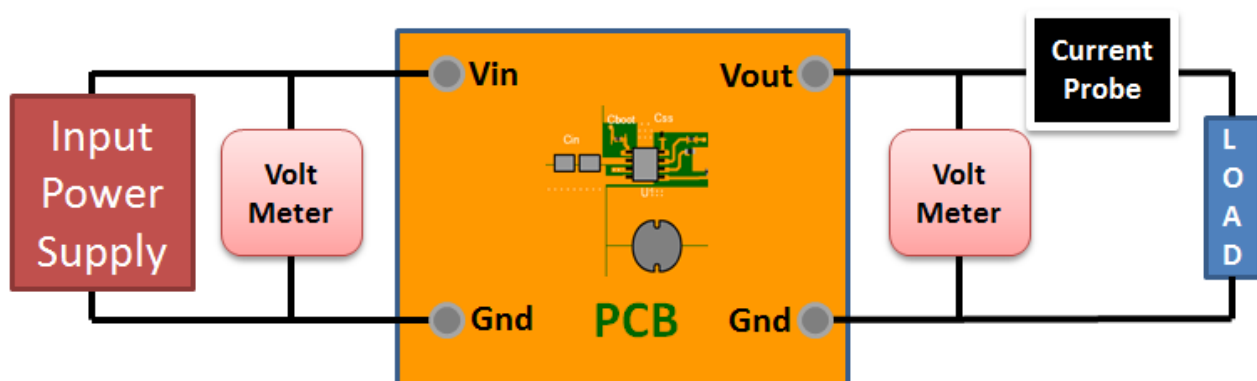
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 19.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : BB521AAEEE95CBF8BC03B55429D1178E[v1]
2. **LM5148** Product Folder : <http://www.ti.com/product/LM5148> : contains the data sheet and other resources.

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